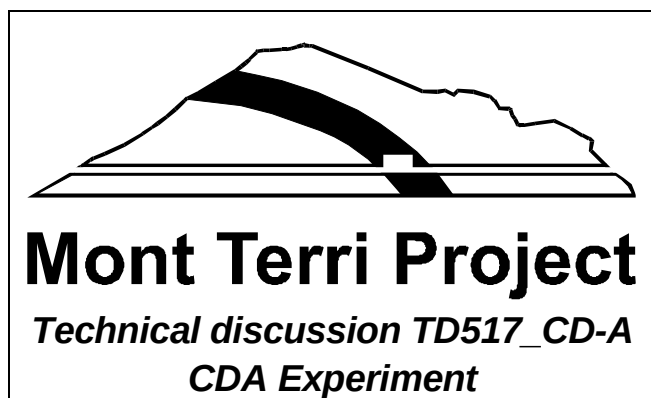




Place: Hybrid Meeting (St. Ursanne, Webex)
Date: April 28 2026, 09:00 – 11:15

<i>Participants</i>		<i>Distribution</i>
BGR: G. Ziefle S. Chen (online) H. Kunz (online) T. Cajuhi (online) S. Costabel (online) M. Furche T. Beilecke V. Kumar GRS: K. Jantschik ISU: F. Königer R. Schuhmann (online) swisstopo: - ENSI: B. Graupner UFZ: - BGE: L. Peti Amphos: - IGN: M. Beres (online)		Participants and D. Jaeggi (swisstopo) K. Rink (UFZ) O. Kolditz (UFZ) J. Maßmann (BGR) T. Burschil (BGR) O. Czaikowski (GRS) J.L. Garcia-Sineriz (Amphos) J. Gerowski (BGR) T. Kneuker (BGR) S. Schefer (swisstopo) S. Condamine (swisstopo) M. Wengler (BGE) S. Tunon Valladares (Amphos) C. Ostertag-Henning (BGR) J. Hesser (BGR) M. Middelhoff (GRS) A. Liebscher (BGE)

Agenda		
Update of measurements (Geoscope and laser scans) has been sent (20.04.26)		
9:00 – 10:30:		
Update of measurements		
1. ERT (Markus) 2. NMR (Stephan) 3. Taupe (Franz) 4. Levelling (Sebastian, Senecio, today: <u>Gesa</u>) 5. Suction (Susana, Jose Luis, today: <u>Gesa</u>) 6. Hannover Meeting focusing on measuring techniques (Taupe, NMR, ERT) 18.09.25		
10:30 – 11:30:		
Modelling activities		
1. Collaborations in the EURAD Hermes Workpackage (Gesa) 2. Modelling at BGR (Shuang)		
Next steps:		
3. Measured trends? Any adjustments needed? 4. Drilling program? 5. Next meeting		
The slides of all presentations of this meeting (Gesa, Markus, Stephan, Franz, Sebastian, Shuang) as well as the current report on suction measurements (Susana) are send by mail (11.05.26).		

1. Update of measurements		
- An update of all measurements that are handled via Geoscope (relative humidity, temperature, opening times of door, extensometer, Mini-Piezometer, crackmeter, suction measurements) plus Laser scans with statistical interpretation has been sent to all partners (Mail sent by Gesa at 20.04.2026) Some remarks to these measurements: - RH and T measurements indicate very cold temperatures and low RH during the winter 25/26 - Data failure of horizontal and vertical extensometers (BCD-A9 and BCD-A10) in the closed niche since September 2025 – next steps? - Mini-Piezometers BCD-A28 to BCD-A31 are working well - Crackmeter indicate ongoing trend in closed niche (1 measuring point) and seasonal variation in open niche 1. ERT Presentation Markus: CD-A_Stand-ERT-2026-04.pdf Update of relation between contact resistances and RH measurements confirms the strong correlation between these values. Comparison of the yearly amplitude indicates a decrease of the amplitude which might be a result of climatic conditions or a general effect. Interpretation of ERT mapping with yearly intervals in terms of relative resistivity changes show no continuous trend of the last 6 years. Resistivity changes are distributed heterogeneous along the niche contour. Largest changes are observed at the top of the niche and at distinct structures like fractures. 2. NMR Presentation Stephan: NMR2026.pdf Weekly measurements of WC in open niche are presented as relationship with RH. While the closed niche behaves as expected (no significant difference), the seasonal changes in the open niche seem to be decreasing. To be observed Further laboratory experiments to interpret the relaxation times of clay rocks have been performed: Drying and		Gesa / All

wetting circles in a climate chamber => NMR at each moisture step. Most important question: Do we see the interlayer water in the Opalinus clay and can we quantify the content using NMR relaxometry? OPA in natural solid formation: bimodal NMR response indicates the two types of water binding. Separating them using cut-off values seem possible, however, diffusion and data noise smears the water signal in the range of the cut-off => quantification still challenging! 3. Taupe Presentation Franz: 2604_TD_CD-A_ISU.pdf (Differential) TDR using TAUPE Sensor Cables: Closed Twin: No significant changes in ARDP during more than 6 years. However, areas with locally increased ARDP can be found near horizontal as well as vertical borehole. Open Twin: Interpretation of the mean amplitude over the first 0.5m: Major variations on ARDP due to meteorological effects, stronger in horizontal direction. A significant reduction in the ARDP from the start of 2025 appears (not at S7front). Interpretation of mean amplitude over varying depth has been performed. Summary: Closed Twin: Horizontal: EDZ narrow, Moderate ARDP in most parts, Medium ARDP around 40 110 cm depth, slightly increasing, Starting increased ARDP zone at rear end Vertical: EDZ quite pronounced, Medium ARDP, in most parts only slightly increasing, Higher ARDP around 150 cm and at borehole end (significant) Open Twin: Horizontal: EDZ pronounced, Slight variations of ARDP between 80 and 300 cm, Significantly higher ARDP behind about 300 cm (Borehole end), Potentially: Wet spots ? Perturbation/fault Vertical: EDZ quite pronounced, Low/medium ARDP over complete borehole, Only slight variations of ARDP over time, Slightly increased ARDP zone at sensor connection -> artificial 4. Levelling Presentation Sebastian/Senecio (<u>Gesa</u>): Folien_Niv_TD_CDA_2026.pdf Precision Levelling Twin Niches Summary: Detectable displacement: from 0.1 to 0.2 mm (depending on the point; Confidence level 95%)	
---	--

Largest deformations: • CDA-O1: -2.1 mm (settlement) • CDA-C4: +1.3 mm (heave / uplift) On all other points: • With the exception of CDA-O1, there is a tendency towards uplift at all points in the Twin Niches. • The height changes in the Open Twin are smaller (except at the aforementioned point O1). Outlook • Measurements over a longer time period • Direct comparison with laser scanning measurements 5. Suction Report Susana/ Jose Luis: RO_2013.4_3974_CD-A3 Phase 30_A21_TN-2025-12_signed.pdf Report summarizes the performance of the suction measuring system, the supervisions, the interventions and gives recommendations for further proceeding Conclusions: The problems related to the functioning of the PSS datalogger persist. The psychrometers software is frequently found stopped and no data being collected. Sometimes this influences the SCADA due to the blockage of the communication port with the RH unit. In general, the problem can be solved by re-starting the PSS software but sometimes the reset of the PSS datalogger is also needed. The data acquisition of the thermo-hygrometers is running smoothly, and the trends are easy to understand. However, data from those installed in the closed twin seem to be affected by the high saturation of the rock driving to an increasing trend above the upper range, which is not realistic and probably related with drift and lack of accuracy at such range. The datalogger of the psychrometers is running with some difficulties but the data provided is usable. In the open niche the ranges are in general below the minimum for them. As conclusion, the thermo-hygrometers are the best for the supervision of the open niche while the psychrometers seems to be better for the closed one. Recommendations: Concerning the thermo-hygrometers system, it is recommended to isolate the connection port from the PSS ones. Initially this could be done with a serial to Ethernet converter. However, as there are two sensors (RH5 & RH7) above from 100% RH, it could happen they finally fail, and this could lead to communication difficulties in the common serial line. This phenomenon was already	
--	--

detected in Sandwich experiment. Thus, the proposal is installing a serial switch instead, which will isolate the serial lines of the 8 sensors and will connect to the computer by Ethernet. Several issues have been identified with the computer, including unexpected restarts and a faulty USB port used for connecting the psychrometer data logger. It is recommended to replace the computer with a new unit running a professional-grade operating system, ensuring compatibility with the legacy controllers required for the psychrometer logger. Amphos is going to provide a proposal for the next steps 6. Hannover meeting on measuring techniques In September 2025 there has been a meeting related to measurements focusing on water content and related values (ERT, NMR, Taupe). It is planned to provide an interpretation of these measurements by means of the water content. The established and updated relationship between resistivity (ERT) and water content (NMR) will be used. An updated version of this relationship is planned for June 2026.	Susana/José Luis Markus/Stephan
2. Modelling	
- BGR Modelling Collaborations (Gesa) There are two collaborations within the EURAD Hermes Workpackage: 1.) Collaboration with IGN (Czech Republic) Optimizing numerical models using geophysical data sets The colleague Tomas Luber (IGN) will be at the BGR from 22.6. to 26.6.26 2.) Collaboration with EDF (France) Validation of modeling results against measurements. 3 step strategy (basic 2D, 2D, 3D) as presented by Shunag - BGR Modelling (Presentation by Shuang): 2D Models -Model with precise surface geometry: solver behavior and efficiency tested -Model with perfect ring geometry: Simulated elastic strain showed discrepancies vs. measurements (extensometer). This is focus of ongoing modeling.	

Modelling Strategy Stepwise benchmarking: Simple 2D → Complex 3D in-situ model. Finally targets to compare with extensometer and pore pressure measurement data. Next steps BGR internal: test of constitutive models. Collaboration: Benchmark BGR vs. EDF modelling.		
3. Various		
Discussion of measurements with focus on measured trends and the question about next steps and possible adjustments of the CD-A experiment: Open twin: ongoing trend can be observed by various types of measurements (extensometer, suction, Taupe...). No adjustments planned. Closed twin: Most of the measurements indicate static conditions (extensometer, pore pressure, suction, ...). The crackmeter measurements at one position in the CT indicate ongoing closure of the crack. Possible adjustments have been discussed and can be further discussed. Current idea/proposal (to be discussed): 2027: Re-Installation of extensometer or fiber optic measurements in the closed twin (failure of old measurements since September 2025) Early 2028: Drilling campaign with geophysical characterization (NMR, ERT, Seismic, Permeability...) End of 2028: Climatization of Open Twin (to be defined)		GRS
4. Next steps		
It is planned to have an online meeting at the 5 th of November at 10am. An invitation will follow.		Gesa
A hybrid TD is planned in combination with the Technical Meeting 2027.		Gesa

Hannover, 11.05.2026

Protocol: Gesa Ziefle, BGR